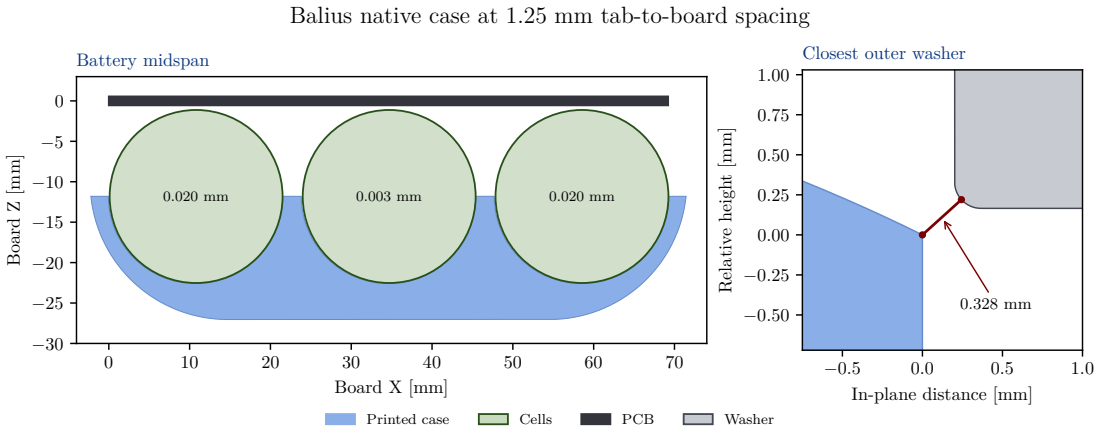
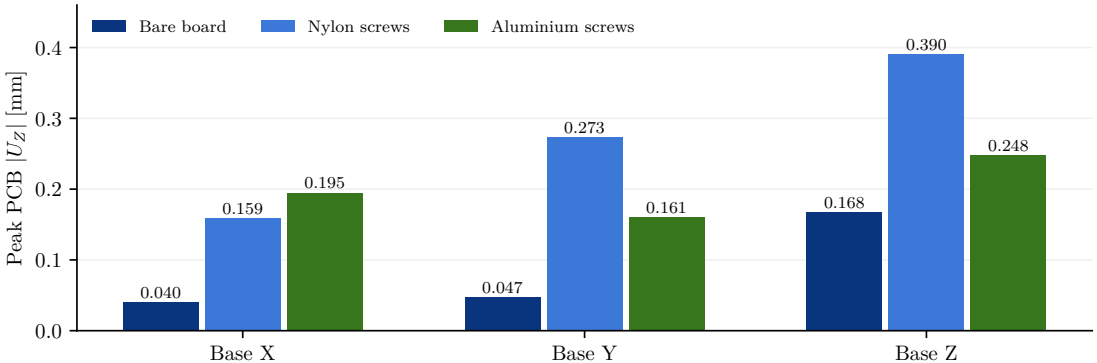
The three-pocket Balius case fits at 1.25 mm stand-off



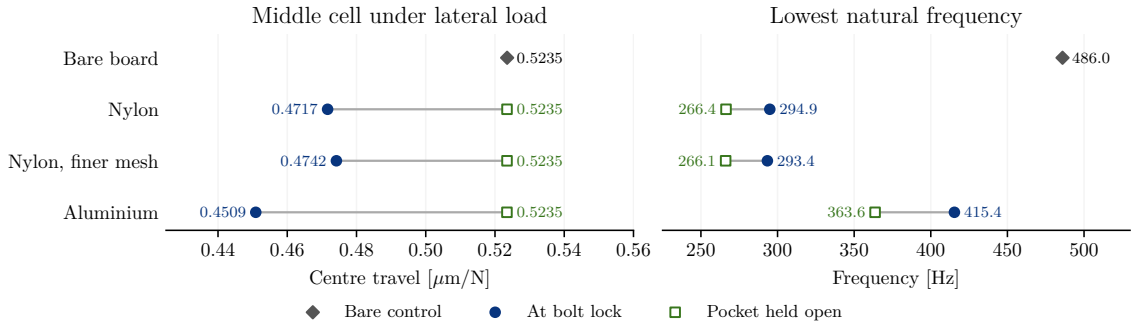
The dense PLA shell adds 73.7 g to the 251.8 g bare board. The middle-cell clearance is 0.003 mm and the closest washer clears by 0.328 mm. The rigid left cell reaches its pocket after **0.020 mm toward the middle cell** or **0.165 mm away from it**.

The snug case increases board travel in the seeded clips



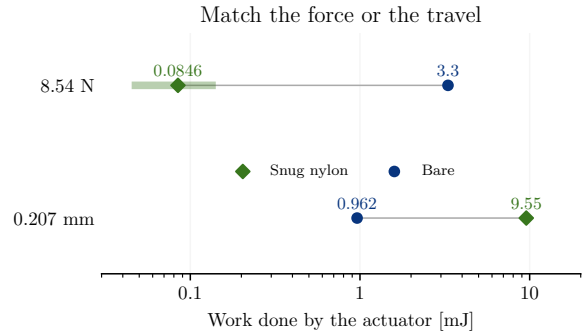
During the displayed base-Z interval, PCB travel normal to the board reached **0.168 mm bare**, **0.390 mm with nylon screws** and **0.248 mm with aluminium**. All 22 paired movie subjects are available.

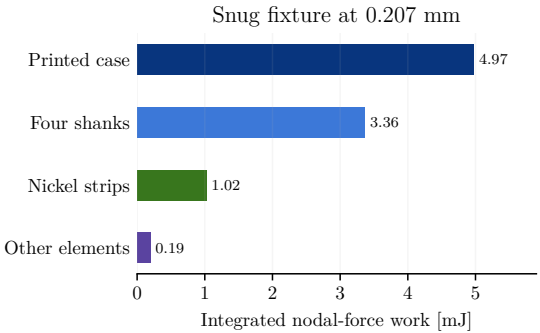
Middle-cell restraint depends on contact with the pocket



At bolt lock, the pocket took **10.4% of a small sideways load with nylon screws** and **14.5% with aluminium**. Holding the pocket open in the whole-board calculation brought cell travel within **0.011% of bare** with either mount. The lowest frequencies also fell, to **266 Hz with nylon** and **364 Hz with aluminium**. The four tabs still attached the case to the board.

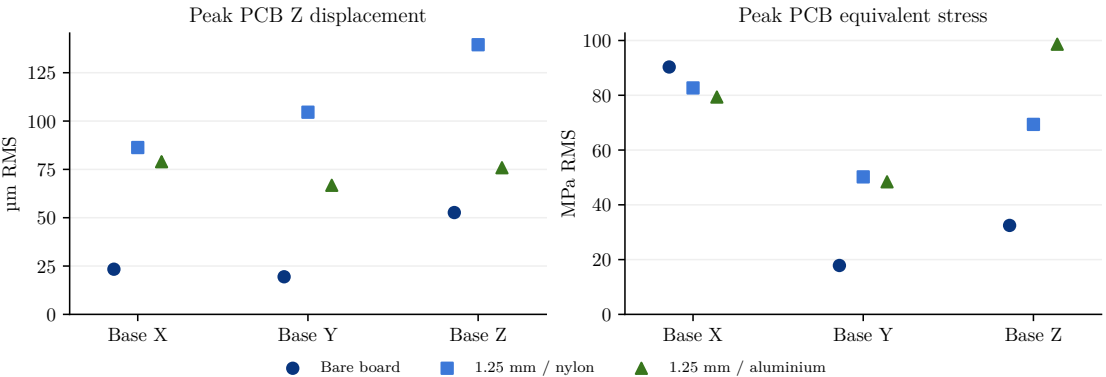
The case reduces edge-cell travel at a given sideways load





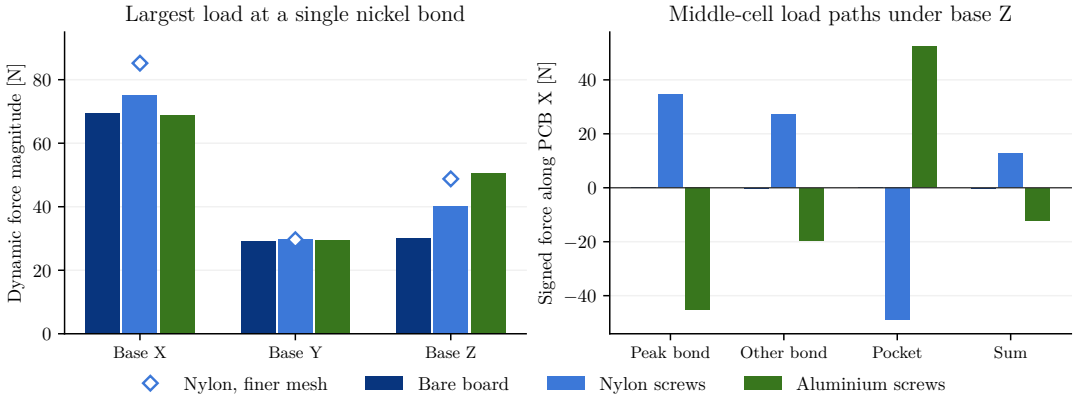
At **8.54 N**, the bare cell travelled **0.500 mm** and the snug nylon cell about **0.034 mm**. Driving both cells to **0.207 mm** took ten times as much work with the case. The PLA and nylon shanks stored **87%** of that work. The fixture holds the neighbours fixed. Refining the tab corner moved its coupon crossing from about 0.30 to 0.23 mm.

Aluminium cuts board travel relative to nylon but raises base-Z stress



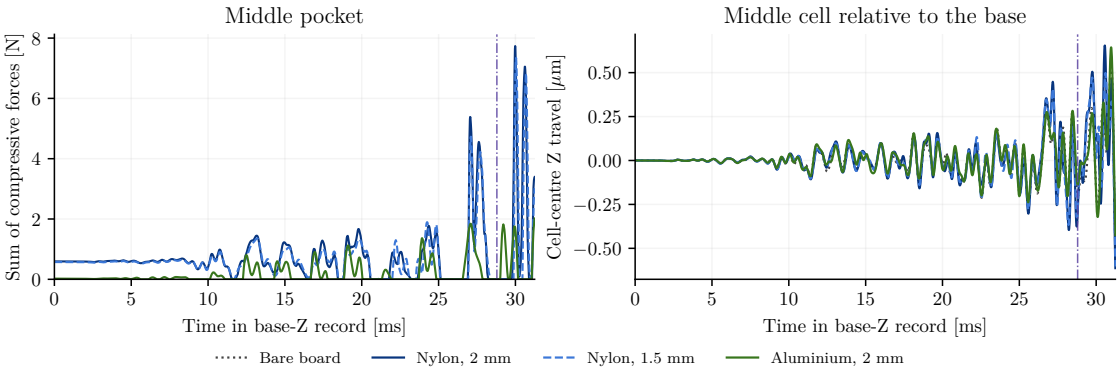
Under base Z, peak PCB stress reached **32.5 MPa RMS bare**, **69.4 with nylon** and **98.7 with aluminium**. Under base Y both mounts raised stress over nearly the entire board. These native spectra hold the installed contact state fixed.

Board-normal vibration loads the middle cell's nickel bond



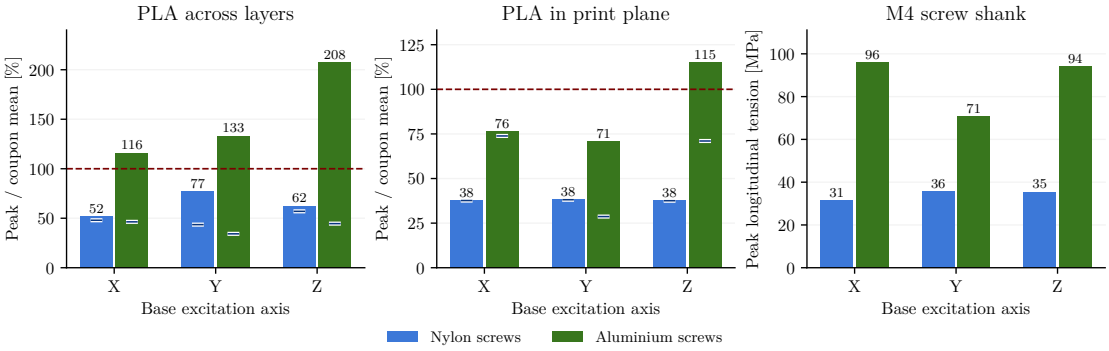
The largest base-Z bond load rose from **30.2 N bare** to **40.1 N with nylon screws** and **50.5 N with aluminium**. The finer nylon mesh gave **48.8 N**. All cased base-Z peaks occur at the middle cell. Bars use native histories at the installed contact state. The diamond uses independent integration on the finer native modes.

The middle pocket unloads and carries force again



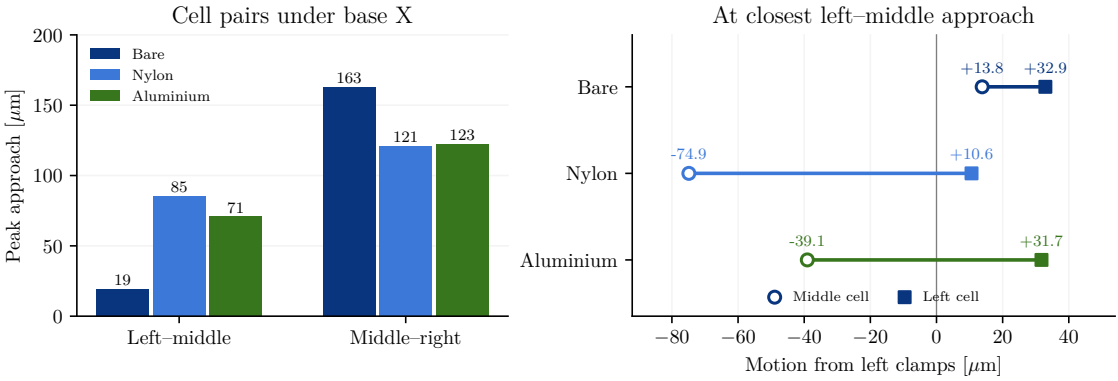
Under base Z, the first interval with all original pocket points unloaded began at **5.61 ms with aluminium** and **11.54 ms with nylon** (11.64 ms on finer nylon). At 28.79 ms the middle cell cleared the case by **3.90–4.01 μm with nylon** and **1.73–1.84 μm with aluminium** on the nominal meshes. New contacts could change this motion.

The printed tab exceeds its coupon reference with aluminium screws



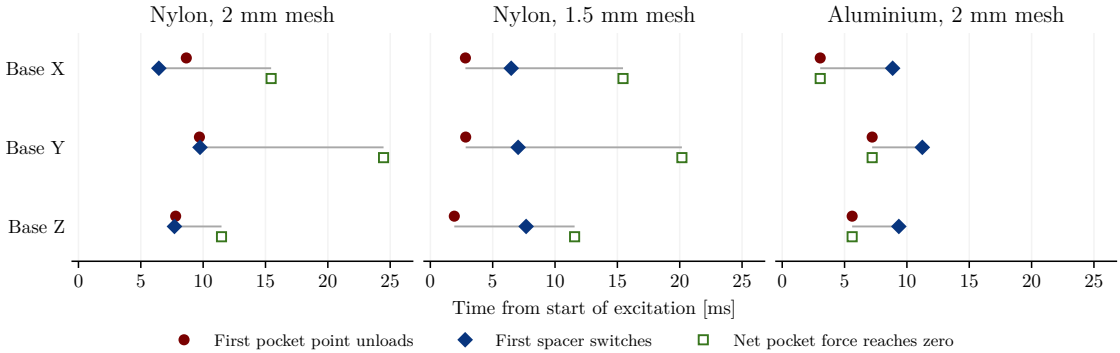
On the nominal mesh the tab reached **77% of the through-layer coupon mean with nylon** and **208% with aluminium**. Refining the nylon case shifted its maximum to **71.5% under base X**. The aluminium mesh check is queued. Horizontal marks exclude the 2 mm bearing band.

Under base X, the left and middle cells move closer together



The left cell's peak travel relative to its clamps fell by **29% with nylon** and **12% with aluminium**. Rotation accounted for **82%** of that travel in all three assemblies. With the middle cell free to move, peak approach rose from **19 μm bare** to **85 and 71 μm**. Contact stays fixed at bolt lock in these native histories.

Which contact changes first during vibration?



With nylon screws, a single lightly loaded pocket point unloaded at **1.92–2.83 ms on the finer mesh**. The middle pocket still carried **0.56–0.57 N**. With aluminium, one point carried the entire **0.0223 N** at bolt lock. Its unloading preceded the first spacer switch under all three inputs. Each marker uses the initial contact stiffness; the motion can change after the first event.